WEALTHWISE

Smart Savings & Tax Planning Platform for India

Project Documentation — Version 4.0

Field	Value
Financial Year	FY 2026-27
Platform	Web App + PWA (Android-installable)
Tech Stack	React 18, FastAPI, Firebase, Claude AI (Anthropic)
Current Version	4.0
Theme	Kuber / Laxmi — Saffron Gold & Temple Emerald
Status	Active Development

1. Executive Summary

WealthWise is an India-specific, AI-powered personal finance web application that helps individuals make smarter savings and investment decisions while minimising their income tax liability under FY 2026-27 rules. Unlike generic financial advisory tools, WealthWise is built from the ground up for the Indian market — incorporating actual IT Act sections, 14 Indian investment instruments, HUF entity support, agricultural income handling, multi-source income aggregation, and the latest Budget 2025 changes including the Section 87A rebate up to ₹12 lakh taxable income.

The platform combines a deterministic multi-income tax engine, a scoring-based risk profiler, a goal-aware portfolio allocator, an IT deductions reference with 20+ sections, a monthly expense tracker with expenditure-based tax deductions, a financial goals SIP planner, and a What-If scenario planner — all integrated with Anthropic Claude AI for plain-English guidance. Deployed as a Progressive Web App (PWA), it is installable on Android without a separate native build.

2. Problem Statement

2.1 The Gap in Existing Solutions

A significant portion of Indian salaried and self-employed individuals lack access to personalised, affordable financial planning. Seven comparable open-source projects were systematically analysed:

Repository	Core Approach	Critical Gaps Identified
Mugeshwaran-k / Financial-Ai-Advisor-Bot	LLM chatbot (Clerk auth)	No Indian tax logic. No real asset data. API key exposed in frontend.
Annarhysa / BudgetBud	Expense tracker with limits	No investment advice. No tax engine. No Indian instruments.
Aaditya-5281 / Finance-Advisor	Generic LLM Q&A	No regime comparison. No risk profiling. No calculations.
marinaredamekhael / FinancialAdvisor	Rule-based budget calc	USD-centric. No India context. No AI. No deductions guide.
hadil1999-creator / RAG_Hack_team	RAG over finance PDFs	Over-engineered. No live data. No portfolio allocation.
imRahul05 / Fin_service	India-focused LLM wrapper	No actual tax slab code. Generic prompting only.
EnigmAI / Financely	ML sentiment + KNN	Non-Indian data. No tax optimisation. API key insecurity.

2.2 Specific Flaws Resolved by WealthWise

- No existing project implements FY 2026-27 Indian income tax slabs, regime comparison, or Section 87A rebate logic in deterministic code — all rely on vague LLM prompting.

- None cover India-specific instruments: PPF, ELSS, NPS, SGB, NSC, SCSS, Sukanya Samriddhi Yojana, or HUF entity-specific tax treatment.
- Five of seven projects expose API keys in frontend code. WealthWise uses a backend proxy — the Anthropic key never leaves the Railway server.
- No project handles multi-source income (salary + rental + FD interest + LTCG + agricultural + crypto) in a single tax computation.
- None implement agricultural income via the partial integration method or loan deductions (home loan, education loan) as deduction sources.
- No project provides expenditure-based tax deductions — WealthWise auto-extracts 80D and 80C hints from the expense tracker.
- No project provides a comprehensive IT Act deductions reference covering 20+ sections with limits, regime applicability, eligibility conditions, and pro tips.
- None implement a mathematically grounded risk profiling algorithm or a goal-aware portfolio adjustment engine.
- RAG pipelines add unnecessary complexity. Claude API with a structured FY 2026-27 system prompt achieves superior accuracy with zero infrastructure overhead.

3. Objectives

1. Build a fully India-specific multi-income tax engine for FY 2026-27 in deterministic code covering salary, rental, FD interest, LTCG, STCG, agricultural income, crypto, and other sources.
2. Support HUF (Hindu Undivided Family) as an entity type alongside Individual, with correct tax treatment.
3. Provide personalised portfolio allocation across 14 Indian and global asset classes with goal-aware adjustment.
4. Show both pre-tax and post-tax CAGR per asset per regime, and dual corpus projections (new regime vs old regime).
5. Create an exhaustive IT Act deductions reference covering 20+ sections with an AI advisor backed by Claude Sonnet.
6. Implement a monthly income and expense tracker with expenditure-based tax deduction extraction (80D, 80C hints).
7. Deliver a financial goals module with automatic SIP calculation and instrument recommendation, integrated into the planner.
8. Build a What-If scenario planner with side-by-side dual projections and milestone snapshots.
9. Ensure production-grade API key security: backend proxy, rate limiting, CORS restriction, spend cap.
10. Deploy as a PWA installable on Android. Support auto savings calculation from tracker data.

4. System Architecture

4.1 High-Level Architecture

```
User (Browser / PWA)        | Vercel CDN  -->  React Frontend  -->  Firebase Auth + Firestore        |
    Backend API (FastAPI on Railway)        | Anthropic Claude Sonnet API
```

4.2 Component Breakdown

Layer	Technology	Responsibility
Frontend	React 18 + Vite	UI, browser-side tax calculations, charts, routing, PWA
Styling	CSS Custom Props + Recharts	Kuber/Laxmi theme (saffron, maroon, emerald), data visualisation
Auth	Firebase Authentication	Google SSO + Email/Password; guest mode (localStorage fallback)
Database	Cloud Firestore	Per-user profiles, transactions, goals (Firebase Security Rules)
Backend	FastAPI (Python 3.11)	AI proxy, multi-income tax API, rate limiting, CORS
AI Layer	Anthropic Claude Sonnet	Portfolio explanations, tax Q&A, savings coaching
Frontend Deploy	Vercel	Global CDN, auto-deploy on git push, env var management
Backend Deploy	Railway	Always-on server, ANTHROPIC_API_KEY stored as env var only

4.3 Security Architecture

- ANTHROPIC_API_KEY stored exclusively in Railway environment variables. Never in frontend code or GitHub repository.
- All AI calls go through /ai/* FastAPI routes. Browser never contacts Anthropic directly.
- Rate limiting via slowapi: 10/min for portfolio explanations, 15/min for tax advisor, 60/min for /analyze.
- CORS locked to production Vercel domain via ALLOWED_ORIGINS environment variable.
- Firebase Security Rules: users can only read/write their own Firestore documents (matched by Firebase UID).
- Monthly spend cap set at console.anthropic.com to limit maximum financial exposure.
- Root cause fix documented: SafeInput and SrcGroup components hoisted to module scope to prevent React remount-on-keystroke bug.

5. Feature Specification

5.1 Multi-Income Tax Engine (FY 2026-27)

The tax engine handles 13 distinct income source types, computes tax under both regimes separately, and returns per-regime marginal slab rates for accurate post-tax corpus projections.

Income Source	Tax Treatment	Special Notes
Salary / CTC	Slab rate	Std deduction ₹75K (new) / ₹50K (old) auto-applied
Business Profit	Slab rate	Net profit after all business expenses
Freelance / Consulting	Slab rate	Net receipts after deductible expenses
Rental Income	Slab rate	30% std deduction under Sec 24 auto-applied
FD / RD / Post Office Interest	Slab rate	TDS @10% above ₹40K/yr (₹50K seniors)
Savings Account Interest	Slab rate	80TTA ₹10K exempt (<60); 80TTB ₹50K exempt (60+); auto from age
Dividend Income	Slab rate	TDS @10% if >₹5K per company
LTCG — Equity / MF	10% above ₹1.25L	FY26-27 exemption limit raised from ₹1L to ₹1.25L
STCG — Equity / MF	15% flat	No exemption. Applies to holdings <1 year.
LTCG — Property / Gold	20% with indexation	Held >2 years. Sec 54/54EC exemption may apply.
Agricultural Income	Exempt + rate computation	Partial integration method: exempt but shifts ordinary income to higher slabs
Crypto / VDA	30% flat + 1% TDS	No deductions, no loss set-off, no exemption. Most restrictive.
Other Income	Slab rate	Pensions, gifts >50K, winnings, etc. (was missing in earlier version — now fixed)

Tax Feature	Detail
New Regime Slabs	0% ₹0-4L | 5% ₹4-8L | 10% ₹8-12L | 15% ₹12-16L | 20% ₹16-20L | 25% ₹20-24L | 30% above ₹24L
Sec 87A Rebate (New)	Zero tax if net taxable ≤ ₹12L (Budget 2025). Gross CTC ≤ ₹12.75L = zero tax.

Tax Feature	Detail
Standard Deduction	₹75K new regime / ₹50K old regime. Auto-applied.
HUF Treatment	Same slabs as individual. No senior citizen benefit. Interest exemption ₹50K.
Loan Deductions (Old)	Home loan interest (Sec 24b, ₹2L cap), education loan interest (Sec 80E, no cap), home loan principal (80C)
Tracker Deductions	Health spending → auto 80D. Investment spending → 80C hint. Pulled from expense tracker.
Per-Regime Slab Rates	NEW: New regime and old regime each get their own marginal slab rate, producing two different post-tax corpora (fixes the identical corpus glitch)
Health & Education Cess	4% on computed tax in both regimes

5.2 Entity Types

Entity	Tax Treatment	Notes
Individual	Standard slabs. Senior benefit at 60+.	Default. Auto-detects senior from age input — no separate toggle needed.
HUF (Hindu Undivided Family)	Same slabs as individual. No senior benefit.	₹50K interest exemption (like senior). Can invest and claim deductions independently.

5.3 Risk Profiler Algorithm

Risk score (0–100) computed from three weighted components, mapped to four profiles:

Component	Scoring Logic	Max Pts
Age	<25: 30 | 25-34: 25 | 35-44: 18 | 45-54: 10 | 55+: 4	30
Occupation	Govt/PSU: 10 | Salaried MNC: 18 | Self-employed/Business: 25	25
Savings Ratio	>=40%: 20 | >=25%: 14 | <25%: 7	20
Profiles	<=35: Conservative | 36-55: Moderate | 56-70: Aggressive | 71+: Very Aggressive	—

5.4 Asset Classes (14 Total) with Tax Rules

Every asset now carries its tax rule. Post-tax CAGR is calculated per regime using that regime's marginal slab rate.

Asset	Pre-Tax CAGR	Tax Rule	Post-Tax CAGR (30% slab)	Lock-in
PPF	7.1%	Tax-Free (EEE)	7.1%	15yr
Savings Account	3.5%	Slab Rate	2.45%	None
Fixed Deposit	7.2%	Slab Rate	5.04%	1-5yr
NPS	11.0%	Partly Tax-Free (60% lump sum)	~9.7%	Till 60
Debt Mutual Fund	7.5%	Slab Rate	5.25%	None
Digital Gold / ETF	11.0%	LTCG 20%	8.8%	None
Sovereign Gold Bond (SGB)	11.0%	Tax-Free (maturity)	11.0%	8yr
ELSS (Tax Saver MF)	14.0%	LTCG 10%	12.6%	3yr
Index MF (Nifty 50)	13.0%	LTCG 10%	11.7%	None
Large Cap MF	14.5%	LTCG 10%	13.05%	None
Mid / Small Cap MF	17.0%	LTCG 10%	15.3%	None
Direct Stocks	15.0%	LTCG 10%	13.5%	None
Real Estate	9.5%	LTCG 20%	7.6%	5yr+
Bitcoin / Crypto	35.0%	30% Flat	24.5%	None

5.5 Goal-Aware Portfolio Adjustment

The Recommended portfolio is automatically adjusted based on the user's financial goals loaded from the Goals tab:

- Goals with timeline ≤3 years: corresponding proportion shifted to FD/Debt for capital safety.
- Goals with timeline 3–7 years: portion shifted to ELSS/PPF for balanced growth.
- Goals with timeline >7 years: full equity allocation maintained.
- Adjustment notes shown on the results page explaining each shift made.
- Other three portfolio options (Safer, Bolder, Tax Optimizer) remain unadjusted for comparison.

5.6 Projection Chart (Corpus Glitch Fix)

Previous versions showed the same corpus for new and old regimes. This was incorrect. The fix:

- New regime and old regime each produce their own marginal slab rate after tax computation.
- Each portfolio asset's post-tax CAGR is computed twice: once with the new regime slab, once with the old regime slab.
- The results page shows THREE lines on the projection chart: Pre-Tax (dashed), New Regime post-tax (emerald), Old Regime post-tax (purple).
- A "tax drag" figure is shown: how many lakhs the user loses to tax over the projection period.

- Users can toggle between 10-year and 20-year projection views (default: 10 years).

5.7 Savings Auto-Calculation from Tracker

Users can toggle between manual savings input and automatic calculation from expense tracker data:

- Manual mode: user types annual savings amount directly.
- Auto mode: reads localStorage transaction data, computes monthly income minus expenses, extrapolates to 12 months.
- If partial year data available: clearly shown as "X months of data — extrapolated to 12 months".
- Monthly savings average displayed alongside the annualised figure for transparency.
- Falls back gracefully with a message if no tracker data exists yet.

5.8 IT Deductions Reference (20+ Sections)

A comprehensive guide where each section includes: exact limit, regime applicability, eligible instruments, conditions, pro tip, and an "Ask AI about this section" button.

Section	Title	Limit	Regime
80C	Investments & Payments	₹1,50,000	Old Only
80CCD(1B)	NPS — Extra Deduction	₹50,000 (over 80C)	Old Only
80CCD(2)	NPS — Employer Contribution	10% salary (Govt: 14%)	Both
80D	Health Insurance Premium	₹25K self + ₹25K parents	Old Only
80E	Education Loan Interest	No upper limit	Old Only
80EEA	Affordable Housing Interest	₹1,50,000 additional	Old Only
80G	Charitable Donations	50%-100% of donation	Old Only
80GG	Rent Paid (No HRA)	₹60,000/year	Old Only
80TTA	Savings Account Interest	₹10,000	Old Only
80TTB	Senior Citizen Interest	₹50,000	Old Only
80DD / 80U	Disability Deductions	₹75K / ₹1.25L severe	Old Only
Section 24(b)	Home Loan Interest	₹2,00,000 self-occupied	Old Only
Section 16	Standard Deduction	₹75K (new) / ₹50K (old)	Both
HRA	House Rent Allowance	Least of 3 formulae	Old Only
LTA	Leave Travel Allowance	Actual travel cost	Old Only
Section 54	LTCG — Property	Full exemption if	Both

Section	Title	Limit	Regime
	Reinvestment	reinvested	
Section 54EC	LTCG — NHAI/REC Bonds	Up to ₹50,00,000	Both
Section 87A	Tax Rebate (New Regime)	Zero tax up to ₹12L taxable	New Only
New Regime Perks	Available in New Regime	Std deduction + employer NPS	New Only

5.9 Expense Tracker

- 16 categories: 11 expense (Rent/EMI, Food, Family, Transport, Utilities, Health/Insurance, Education, Shopping, Investments, Insurance Premiums, Other), 5 income.
- NEW: Family category added for household/dependent expenses.
- NEW: Expenditure-based tax deductions panel — health spending auto-maps to 80D, investment spending flags as potential 80C. Shown with section badge and deductible amount.
- Deductions from tracker auto-applied in the Tax Planner (old regime) without manual entry.
- Month/year selector + all-time view. KPI bar: income, expenses, net savings, savings rate %.
- Category breakdown pie chart, 6-month income vs expenses bar chart.
- AI Savings Coach: sends spending summary to Claude via backend for personalised tips.
- Firestore sync for authenticated users; localStorage fallback for guests.

5.10 Financial Goals with SIP Calculator

- 8 pre-built templates: House, Car, Education, Wedding, Travel, Retirement, Emergency Fund, Business.
- Custom goal: name, emoji, target amount, timeline (years), category, amount already saved.
- Live SIP preview: monthly amount required + recommended instrument before saving.
- Instrument recommendation: maps goal timeline to optimal asset (e.g. FD for <1yr, Index MF for 7yr+).
- SIP formula: SIP = (Target − Saved) × r / ((1+r)^n − 1) where r = monthly rate, n = months.
- Goals loaded into Planner tab: portfolio automatically adjusted based on goal timelines.
- Progress bar, percentage funded tracker, update anytime. Firestore sync / localStorage fallback.

5.11 What-If Scenario Planner

- Side-by-side comparison of two fully customisable investment scenarios.
- 4 presets: Early vs Late Start, SIP Step-up Effect, Conservative vs Aggressive, Save More vs Invest Smarter.
- Inputs: annual savings, CAGR %, years of investment, annual step-up %.
- Dual-area projection chart, difference callout, milestone table at years 5/10/15/20/25/30.

6. Technical Implementation

6.1 AI Integration Design

Approach	Build Time	Accuracy	Cost	Chosen?
Train a custom model	3-6 months	Low — needs massive labelled data	GPU servers	No
RAG pipeline (vector DB)	2-3 weeks	Good	Vector DB + embedding API	No
Claude API + system prompt	1 day	Excellent — native FY26-27 knowledge	Pay-per-call	YES

RAG was explicitly rejected. IT Act content is structured, static, and within Claude's training distribution. Hardcoding FY 2026-27 rules into the system prompt achieves superior accuracy with zero infrastructure. The sole future exception: Form 16/AIS PDF upload, where Claude's native document API would be used directly — still no RAG required.

6.2 Frontend Architecture

File	Purpose
App.jsx	Main app: 5-page routing, all planner steps, bottom nav, savings toggle, goal loading
IncomeForm.jsx	Multi-source income entry. SafeInput and SrcGroup hoisted to module scope (cursor/scroll fix).
taxEngine.js	All tax logic, risk profiler, portfolio generator, projection, tracker savings calc. Pure JS.
api.js	All backend HTTP calls. No API key present anywhere in this file.
Login.jsx	Google Sign-in + Email/Password UI with guest skip option.
AuthContext.jsx	Global auth state via React Context. Auto-loads Firestore profile on login.
firebase.js	Firebase init + all Firestore CRUD helpers for profiles, transactions, goals.
ITSections.jsx	20+ IT Act sections guide. Per-section AI advisor calls /ai/tax-advisor on backend.
Tracker.jsx	Expense tracker with Family category, expenditure deductions panel, AI savings coach.
Goals.jsx	Goal setting with SIP calculator, instrument recommendation, progress tracking.
Scenarios.jsx	What-if planner with 4 presets, dual chart, milestone

File	Purpose
	table.
index.css	Kuber/Laxmi theme: saffron gold (#E8921A), midnight maroon (#0e0806), temple emerald (#1DB873).
main.jsx	Entry point. Wraps app in AuthProvider.

6.3 Backend API Routes

Route	Method	Rate Limit	Description
GET /health	GET	None	Server health check
GET /tax-multi	GET	None	Quick multi-income tax breakdown for testing
POST /analyze	POST	60/min	Full analysis: multi-income tax + risk + portfolios + projections (10yr default)
POST /ai/explain-portfolio	POST	10/min	Claude explains portfolio with post-tax CAGR and income mix context
POST /ai/tax-advisor	POST	15/min	Free-form tax Q&A. Includes user profile context (income, slab, occupation).
GET /docs	GET	None	Auto-generated Swagger API documentation

6.4 Key Bug Fixed in v4 — IncomeForm Cursor & Scroll

Both the cursor-disappears-after-one-digit bug and the scroll-to-top-on-keypress bug shared the same root cause:

- SafeInput and SrcGroup were defined inside IncomeForm. On every keystroke, the parent re-rendered, creating new function references. React saw new component types, unmounted the old input, and mounted a fresh one — losing focus after exactly 1 digit.
- The same re-render caused a layout recalculation that scrolled the page back to the top.
- Fix: both components hoisted to module scope. Their identity is now stable across parent re-renders. React reuses the same instances and never unmounts the input mid-typing.
- Additional: taxResult debounced 400ms so the live tax preview does not recalculate on every keypress.
- Additional: card onClick guard — if (e.target.tagName === 'INPUT') return — prevents the card toggle from firing when typing inside it.

7. Deployment

7.1 Infrastructure & Cost

Service	Provider	Tier	Monthly Cost
Frontend hosting	Vercel	Hobby (free)	₹0
Backend hosting	Railway	Hobby ($5 credit)	₹0
Database + Auth	Firebase (Spark Plan)	Free tier	₹0
AI API	Anthropic	Pay-per-use	~₹0.50 per 100 AI calls
Domain (optional)	Namecheap (.in)	Annual subscription	~₹67/month
Total	—	—	₹0–₹67/month

7.2 CI/CD Pipeline

11. Developer edits code and tests locally on localhost:3000 (frontend) and localhost:8000 (backend).
12. Commit and push: git add . && git commit -m "description" && git push
13. Vercel auto-detects the push, rebuilds React app, deploys to CDN (2-3 minutes).
14. Railway auto-detects the push, rebuilds FastAPI server, restarts (2-3 minutes).
15. Zero manual steps. Changes are live at the production domain automatically.

7.3 Environment Variables

Variable	Location	Description
ANTHROPIC_API_KEY	Backend — Railway	Claude API key. Never in frontend or repository.
ALLOWED_ORIGINS	Backend — Railway	CORS whitelist. Vercel domain only in production.
VITE_API_URL	Frontend — Vercel	Backend base URL for all AI proxy calls.
VITE_FIREBASE_API_KEY	Frontend — Vercel	Firebase project identifier (safe to expose).
VITE_FIREBASE_AUTH_DOMAIN	Frontend — Vercel	Firebase auth domain for OAuth redirect.
VITE_FIREBASE_PROJECT_ID	Frontend — Vercel	Firebase project ID.
VITE_FIREBASE_STORAGE_BUCKET	Frontend — Vercel	Firebase storage bucket.
VITE_FIREBASE_MESSAGING_SENDER_ID	Frontend — Vercel	Firebase Cloud Messaging sender ID.
VITE_FIREBASE_APP_ID	Frontend — Vercel	Firebase web app identifier.

8. Comparative Analysis

Feature	WealthWise v4	Best Competitor
India FY 2026-27 tax engine	Full code implementation	None of 7 projects
New vs Old regime comparison	Yes — with rupee savings + per-regime slab rates	None
Section 87A rebate (Budget 2025)	Implemented	None
HUF entity type	Yes — correct tax treatment	None
Agricultural income (partial integration)	Yes	None
Multi-source income (13 types)	Yes — salary, rental, LTCG, crypto, agri, other, etc.	None
Loan deductions (home, education)	Yes — old regime	None
Per-asset post-tax CAGR	Yes — per regime	None
Dual corpus chart (new vs old)	Yes — 3 lines (pre-tax, new, old)	None
20+ IT Act sections + AI advisor	Complete guide	None
14 Indian + global asset classes	PPF, NPS, ELSS, SGB, Crypto, Savings A/C, etc.	None
Expenditure-based tax deductions	80D from health spending, 80C hints	None
Goal-aware portfolio adjustment	Yes — adjusts allocation by timeline	None
Savings auto-calc from tracker	Yes — toggle manual/auto	None
Mathematical risk profiling	3-factor scoring algorithm	Questionnaire only (Financely)
API key security (backend proxy)	Full — key never in frontend	Mostly exposed in frontend
Firebase Auth (Google + Email)	Yes + guest mode	Mugeshwaran-k (Clerk only)
Expense tracker with AI coach	Yes + family category + tax deduction panel	Annarhysa/BudgetBud (no AI)
Goals + SIP + instrument recommendation	Yes + integrated into planner	BudgetBud (basic)
Scenario planner with milestones	Yes — 4 presets + custom + dual chart	imRahul05 (basic)
PWA / Android installable	Yes (manifest.json)	None
ML model training required	No	EnigmAI (KNN + K-Means)
RAG pipeline required	No	hadil1999 (vector DB + embeddings)
Kuber/Laxmi visual theme	Yes — saffron, maroon, emerald	None

9. Limitations & Future Work

9.1 Known Limitations

- Asset CAGR values are static historical averages — not connected to live AMFI, MCX, or CoinGecko data.
- HRA, LTA, and actual home loan interest amounts are not individually input in Step 1 yet (standard amounts assumed for old regime comparison).
- No Form 16, AIS, or investment statement PDF upload/parsing.
- Futures and options are excluded from portfolio allocation (require specialist knowledge to recommend safely).
- React Native / Expo app for Play Store not yet built; currently PWA-only for Android.

9.2 Roadmap

Planned Feature	Priority	Effort
Live MF NAV data via AMFI API	High	1 week
Live FD rates via RBI / bank scraping	High	1 week
HRA + LTA inputs in tax engine Step 1	High	3 days
Form 16 / AIS PDF upload + parsing (Claude doc API)	High	2 weeks
Crypto tax calculator (30% flat + 1% TDS per tx)	Medium	3 days
Live stock/MF prices tab (yfinance + AMFI)	Medium	2 weeks
n8n automation for daily rate updates	Medium	1 week
Open-source LLM option (Groq / Gemini free tier)	Medium	3 days
CSV / PDF export of full financial plan	Medium	1 day
SIP reminder push notifications (PWA)	Medium	1 week
React Native app via Expo EAS for Play Store	Medium	3 weeks
PyPortfolioOpt (Markowitz MPT) for v5 portfolio	Low	4 weeks
Multi-language support (Hindi, Tamil, Telugu)	Low	3 weeks
DigitalOcean migration + Mumbai CDN for India users	Low	1 week

10. References

10.1 Competing Projects Analysed

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End of Document — WealthWise Project Documentation v4.0